Open Babel has out-of-bounds write in MOPAC translationVectors[] (FINAL POINT)

Report Type: Weekly · Generated: July 01, 2026 18:16 UTC · Coverage: Jul 01 - Jul 01, 2026

1 Total Advisories	0 Critical	1 High	0 Medium
HIGH Severity	7.8 CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Summary A memory-safety vulnerability in Open Babel's MOPAC output parser allowed an out-of-bounds write into the `translationVectors[]` array when reading the "FINAL POINT" block of a crafted input file. ### Details The MOPAC output reader stored translation vectors from the FINAL POINT block into a fixed-size `translationVectors[]` array. A malformed block could push more vectors than the array had slots, causing a write past the end of the array. One of five `translationVectors[]` OOB writes in the TALOS 2022 batch. ### Impact Open Babel is a C++ library and CLI used to read and write chemistry file formats; it is shipped by Linux distributions and embedded in services that may parse untrusted input. Triggering this vulnerability requires the victim to open a malicious MOPAC output file with the `obabel` tool, the `OBConversion` API, or any of the language bindings (Python, Ruby, Java, R, Perl, C#, PHP). ### Affected versions All releases up to and including 3.1.1. ### P

Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
Open Babel has out-of-bounds write in MOPAC translationVectors[] (FNA	HIGH	7.8	82.0%	### Summary A memory-safety vulnerability in Open Babel's MOPAC output par allowed an out-of-bounds write into the `translatio

MITRE ATT&CK® v15 Mapping

Technique density score: **0.0** · Techniques observed: **0** · Tactics covered: **0**

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - Heap Overflow Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: |
Detects exploitation attempts targeting CDB-ADVISORY (heap-based overflow).
Monitors for anomalous process crashes and memory allocation failures.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/01
tags:
- attack.initial_access
- attack.t1190
- attack.execution
- attack.t1203
logsource:
category: process_creation
product: windows
detection:
selection_crash:
EventID: 1000
ApplicationName|contains|any:
- 'Open'
selection_wer:
EventID: 1001
AppPath|contains|any:
- 'Open'
condition: selection_crash OR selection_wer
falsepositives:
- Legitimate software bugs / crash reporting
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - Open Babel has out-of-bounds write in MOPAC translationVectors[] (FINAL POINT)
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "Open Babel has out-of-bounds write in MO"
or AlertName has "Open Babel has out-of-bounds write in MO"
or ExtendedProperties has "Open Babel has out-of-bounds write in MO"
| extend Rule = "APEX-ADVISORY", Severity = "HIGH"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield Open Babel has out instances exposed to the internet.
3. Enable verbose access logging on all Open Babel has out endpoints immediately.
4. Pull last 72h of web server and application logs for forensic baselining.
5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

1. Apply vendor patch for Open Babel has out or implement recommended mitigation.
2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
3. Rotate all API keys, service account credentials, and session tokens.
4. Conduct forensic timeline reconstruction for potential compromise window.
5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

1. Conduct post-incident review and update incident response runbooks.
2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
3. Update asset inventory with patched version information and closure evidence.
4. Submit findings to internal risk register and CISO dashboard.
5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

$800K - $4.5M	**$1.8M**	**$450K/day**	**$320K**
Breach Cost Range (FAIR)	IBM Cost of Breach 2025 Median	Business Interruption Cost	Regulatory Fine Exposure

High severity incidents activate IR retainers, forensic investigation (avg 47 days), and reputational damage impacting 12-18% of revenue.

Remediation Cost Estimate: $180K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
GDPR	Art. 32 / Art. 33	Appropriate technical measures; notify DPA if personal data affected
PCI-DSS v4.0	Req. 6.3.3	Patches within 1 month; documented risk acceptance if deferred
NIS2 Directive	Art. 21	Risk management measures including vulnerability handling policies
NIST CSF 2.0	RS.MI / RC.RP	Incident mitigation and recovery planning; post-incident review

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for Open Babel has out-of-bounds write in MOPAC translationVectors[] (FINAL POINT).
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK techniques.
4. Apply vendor patch for Open Babel has out-of-bounds write in MOPAC translationVectors[] (FINAL POINT) immediately - check NVD for official fix guidance.
5. Enable enhanced logging and alerting on all systems running affected software versions.
6. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
7. Audit all internet-facing instances of affected products and confirm patch deployment.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--9ded39bedc25546d71fbb723
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-01T18:16:38.41859
Customer Tier	PRO
Customer Email	-
TLP	TLP:CLEAR
STIX Version	2.1
ATT&CK Version	v15

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